

DVD Rental Analytics Executive Report

Overview

This report presents findings from a full analytics review of the DVD rental business. It covers customer value and retention, revenue performance across film categories, inventory utilisation, and store operations. The goal is to give leadership a clear, evidence-based picture of where the business stands and where the biggest opportunities lie.

Summaries

599 Total customers	16,044 Total rentals	61,312 Total revenue	2 Stores
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1. Customer Analytics

Understanding who our customers are and how much value they bring is the foundation of any retention strategy. This section covers how customers are distributed by spend, what their lifetime value looks like, and which customers are at risk of not coming back.

Customer segmentation

Customers were split into three equal-sized spend tiers based on total amount paid. The high segment accounts for 26,010 in revenue, nearly double the low segment..

Segment	Customers	Avg Spend	Total Revenue	Spend Range
Low	200	76.65	15,330	27.93 - 89.77
Mid	199	100.37	19,973	89.78 - 110.72
High	200	130.05	26,010	110.73 - 211.55

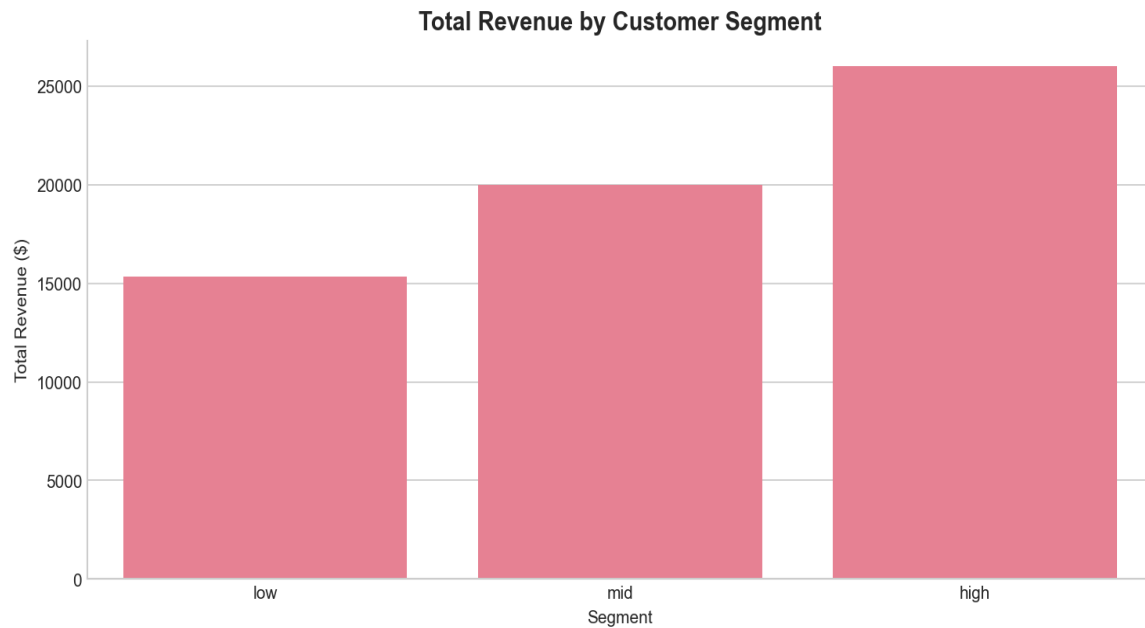


Figure 1. Total revenue by customer segment

Customer lifetime value

CLV was calculated using average purchase value, purchase frequency, and customer lifespan. The top customers show CLV values above 90, driven by a combination of higher spend per transaction and longer active periods. The distribution is right-skewed, with most customers clustering between 15 and 40 CLV, and a smaller group of high-value customers well above that range.

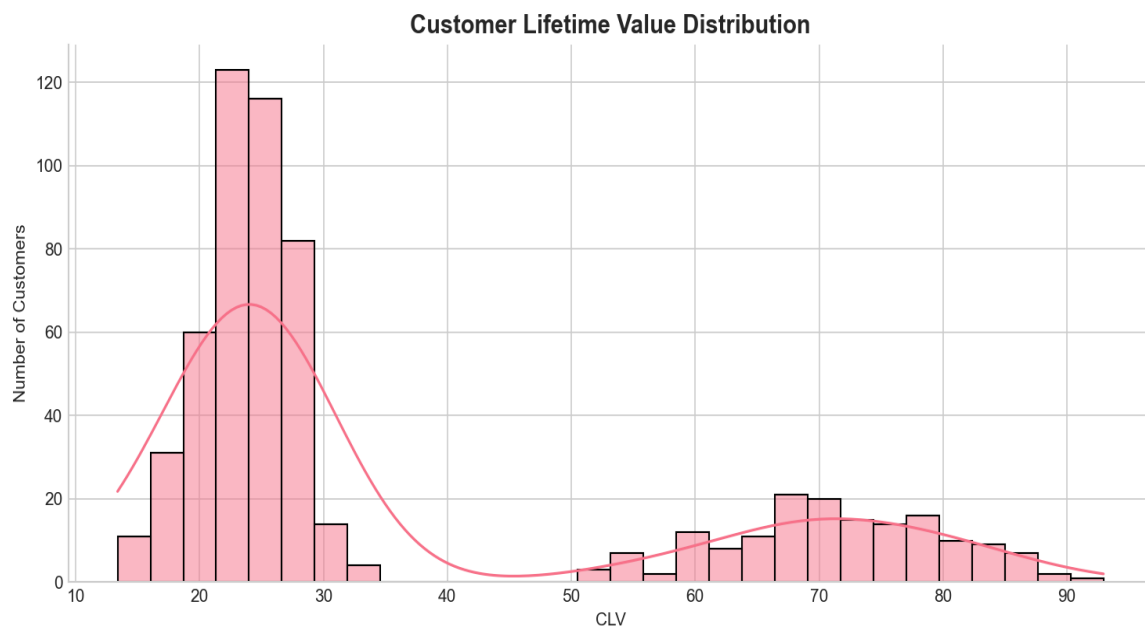


Figure 2. Distribution of customer lifetime value

Churn risk

Using last rental date as a proxy for activity, customers were classified into active (within 30 days), at-risk (31-90 days), and churned (90+ days) relative to the most recent transaction in the dataset. 158 customers are active, 441 are classified as churned. The high churn count reflects the snapshot nature of the data, where most activity was concentrated in a narrow window.



Figure 3. Churn risk distribution across the customer base

Return behaviour

Payment rate measures how many of a customer's rentals resulted in a recorded payment. Most customers have a rate close to 1.0, meaning nearly all rentals were paid. A small number of customers show rates below 0.8, indicating outstanding returns that represent unrecovered revenue.

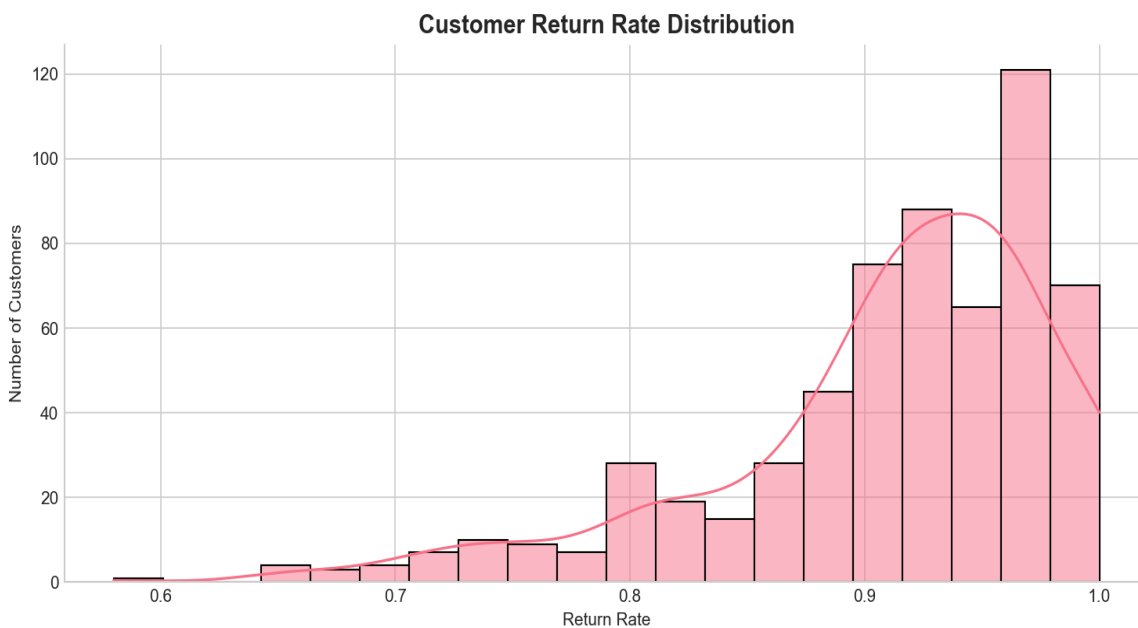


Figure 4. Customer return rate distribution

2. Revenue Optimisation

Revenue analysis focuses on which film categories are performing, whether popularity translates to earnings, how demand shifts over time, and whether pricing differences explain revenue gaps between categories.

Category performance

Sports ranks first in both revenue (4,892) and rental volume (1,179 rentals), making it the strongest overall category. Comedy and New are ranked 5th and 6th by revenue but sit at 11th and 12th by volume, suggesting these categories command a higher price per rental relative to how often they are rented. Action is the third most rented category but ranks 7th by revenue, indicating it may be underpriced. Family shows a similar pattern, ranking 5th by volume but 10th by revenue.

Payment rate by category was included to test whether low revenue in high-volume categories could be explained by customers not returning films. Across all categories the payment rate sits between 0.89 and 0.92, meaning the gap between rentals and payments is consistent and not category-specific.

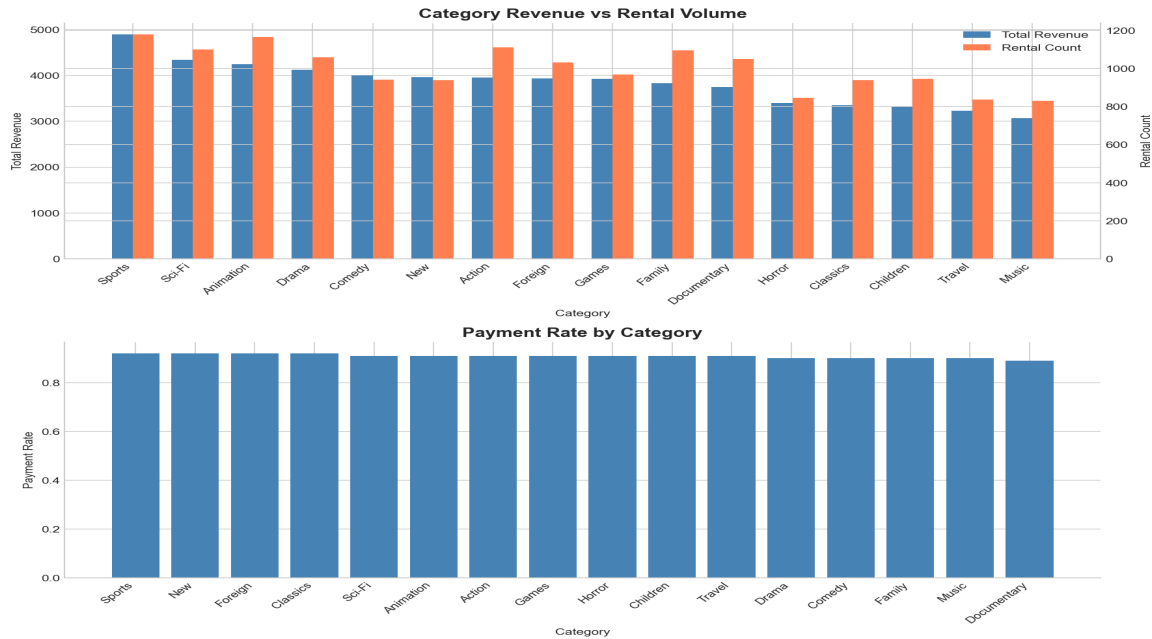


Figure 5. Category revenue rank vs rental volume rank, with payment rate by category

Seasonal trends

This was to understand if seasonality can be a result of client rental behaviour. Rental activity peaks in July and August, with Sports, Animation, and Action consistently in the top three. Animation holds a top-two position every month it appears, suggesting stable demand rather than seasonal spikes. The February data point is a small sample and should not be used for seasonal conclusions.

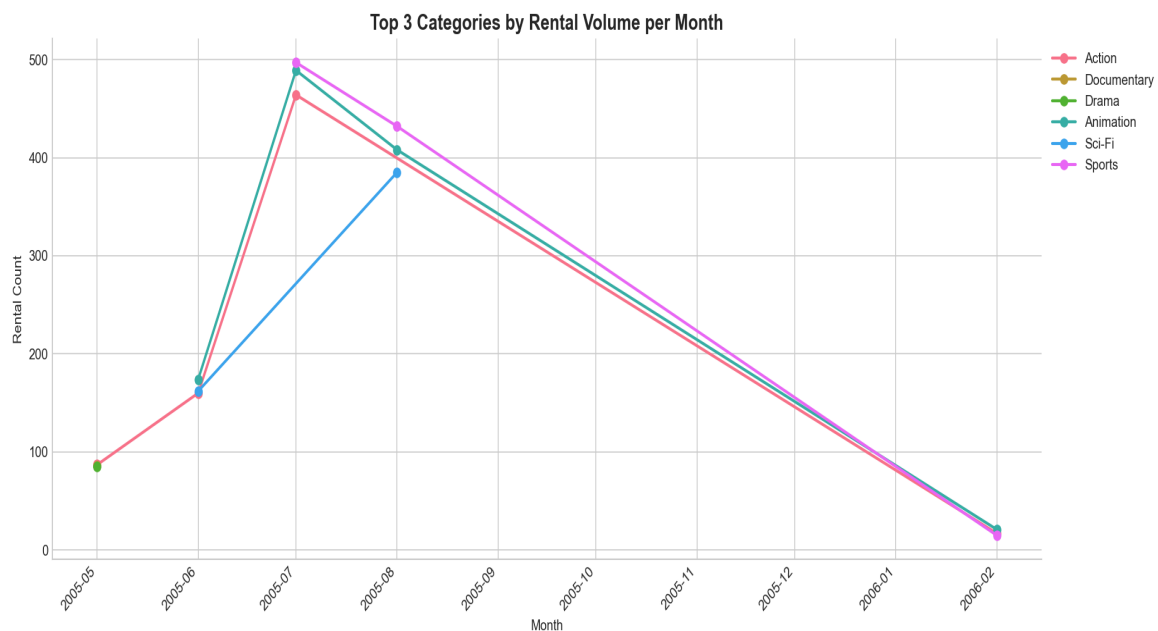


Figure 6. Top 3 categories by rental volume per month

Pricing insights

This was done to understand if pricing could be the reason some categories are performing relatively poorly. Average payment per rental varies across categories. Comedy, New, and Horror show higher average payments relative to their rental volume, while Action, Family, and Children show lower average payments despite strong demand. This suggests an opportunity to review pricing on high-volume, low-revenue-per-rental categories to close the gap between popularity and earnings and also to assess whether high-priced, low-volume categories could attract more demand with a price adjustment.

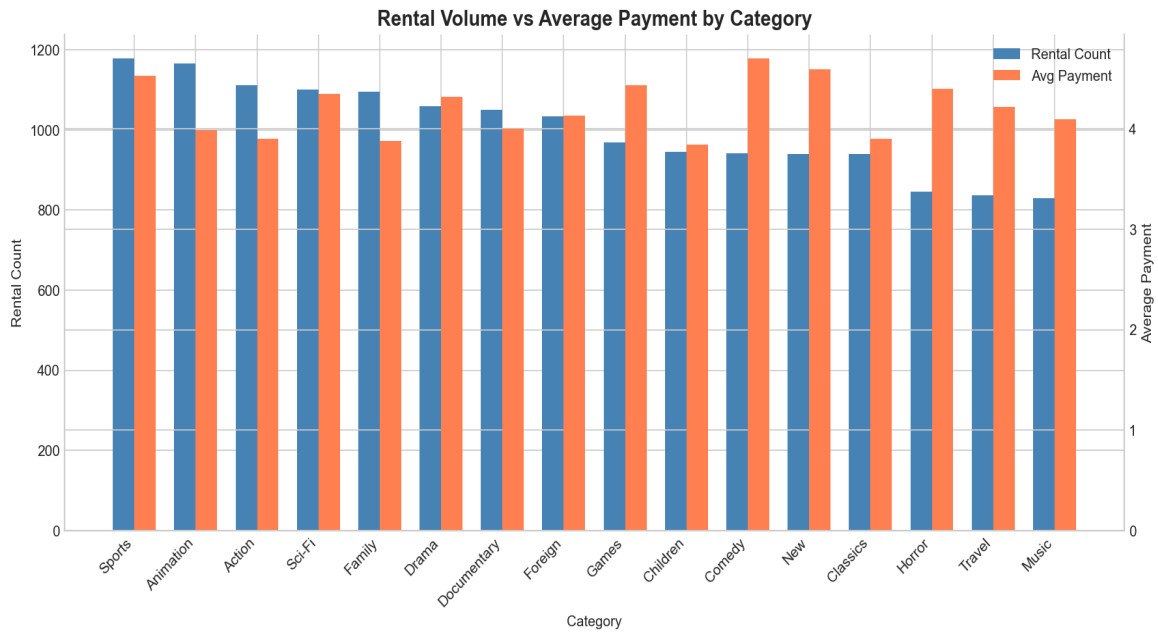


Figure 7. Average payment amount vs rental volume by category

3. Inventory and Operations

Efficient inventory management means matching stock levels to demand. This section identifies which films are turning over quickly, which are sitting idle, and how the two stores compare in performance and category mix.

Film turnover

Turnover rate is calculated as total rentals divided by number of copies. A rate of 5.0 means each copy was rented five times on average. The top-performing films include Anthem Luke, Antitrust Tomatoes, and Blood Argonauts, all with a turnover rate of 5.0. These titles are generating maximum return from their available copies.

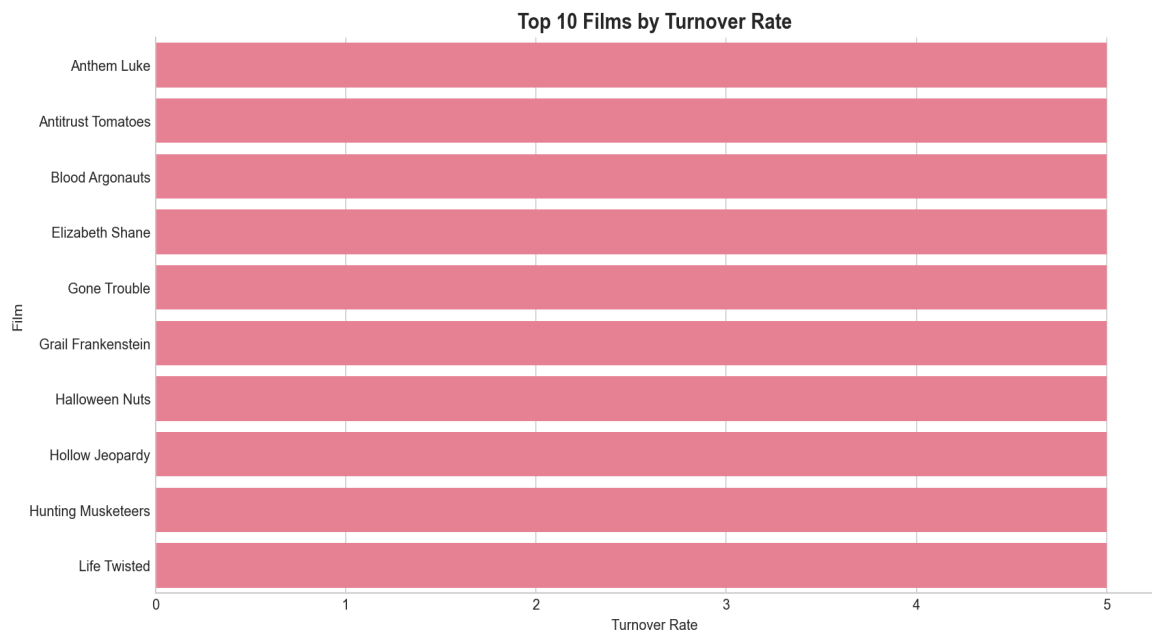


Figure 8. Top 10 films by turnover rate

Stock efficiency

Overstocked films are those with 8 copies but low rentals per copy. Academy Dinosaur has 8 copies but only 2.88 rentals per copy. Understocked films have only 2 copies but high demand, with rentals per copy reaching 5.0. Redistributing copies from low-demand to high-demand titles would improve overall utilisation without additional stock investment.

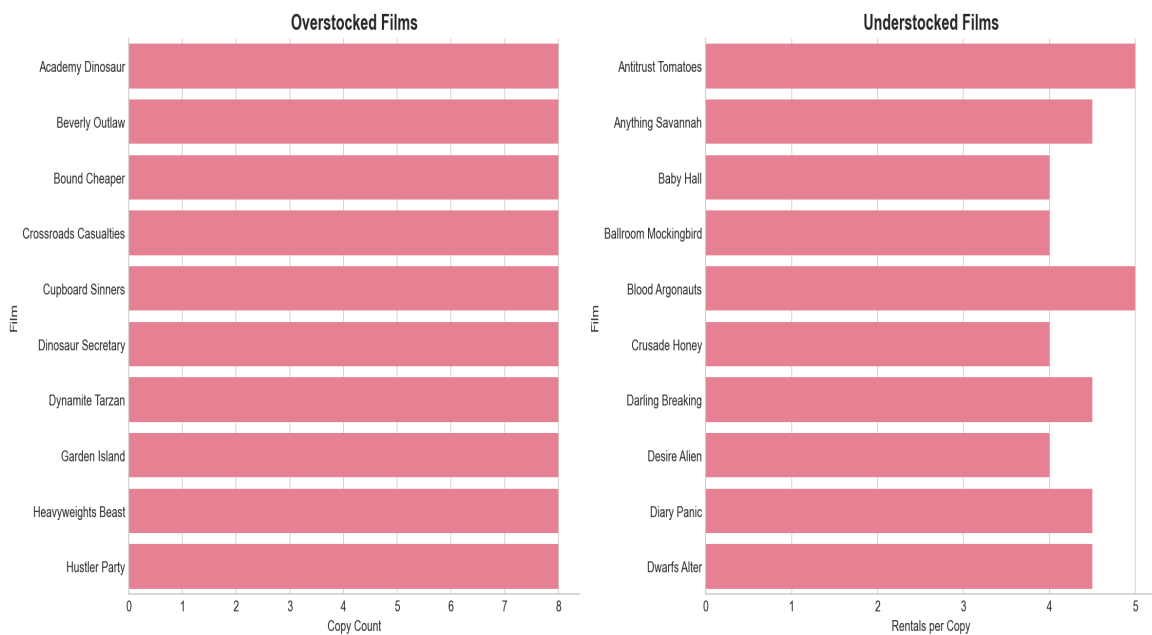


Figure 9. Overstocked and understocked films by rentals per copy

Store comparison

Store 2 processed 8,121 rentals versus 7,923 at Store 1, a difference of 198 rentals. The gap is modest and both stores share a similar category mix, with Sports, Animation, and Action among the top categories at each location. The performance difference is unlikely to be driven by category selection alone and may reflect foot traffic, location, or staffing factors outside the dataset.

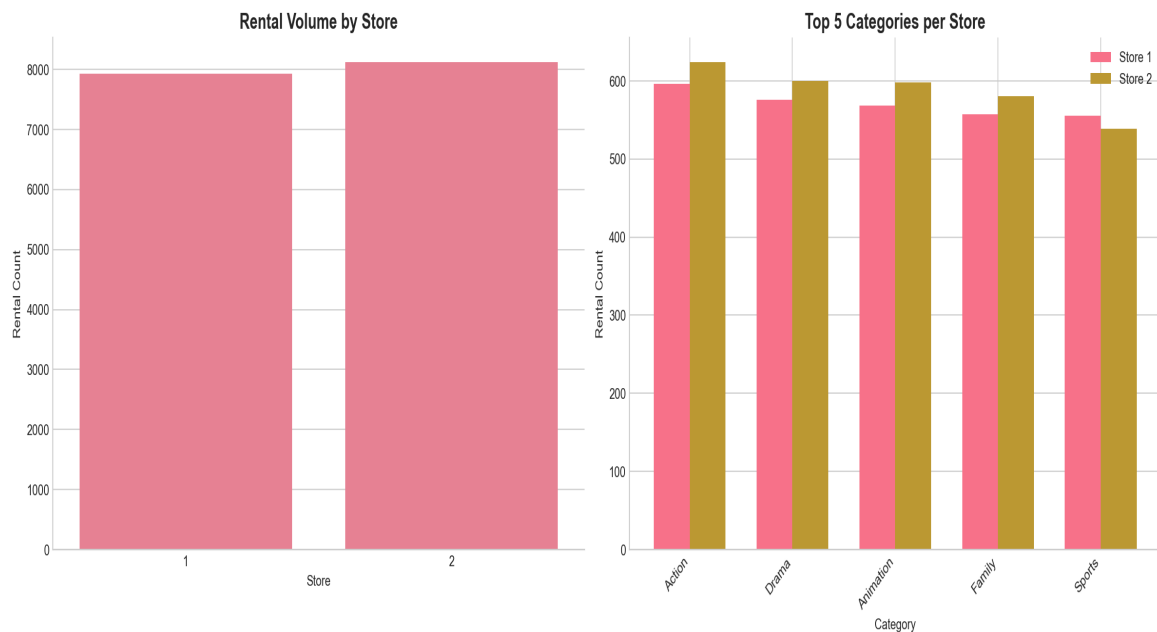


Figure 10. Rental volume by store and top categories per store

4. Churn prediction

This was to understand which customers are likely to stop renting before they actually do which gives the business a window to act. This section covers what we attempted to build and what it would take to make it useful.

What we attempted

Using rental frequency, payment rate, customer lifespan, and lifetime value as inputs, we trained two classification models to predict whether a customer would be active, at-risk, or churned.

Both models achieved perfect accuracy across all folds. While that sounds promising, it is actually a signal that the model is not learning anything meaningful. The features used to train the model are derived from the same transaction data used to define churn, so the model is effectively memorising the labels rather than discovering patterns that would generalise to new customers.

Business impact

The result tells us that rental frequency, payment rate, and spend are the right variables to use and they carry enough signal to separate active from churned customers cleanly. What is missing is the data volume and time depth needed to turn that signal into a forward-looking prediction rather than a retrospective classification.

What would make it viable

A reliable churn prediction model would require transaction history spanning at least two to three years, a larger customer base, and ideally some record of customers who were contacted and either retained or lost. With that data, the same approach would produce a model that flags at-risk customers before they churn, enabling the business to direct retention spend where it is most likely to have an impact.

5. Innovation: Market Basket Analysis

The temporal trends analysis tells us which top 3 categories perform best each month. What it does not tell us is what else customers are renting as a result of getting those films. A market basket analysis on rental history would answer that question.

The opportunity

By looking at which categories tend to appear together in a customer's rental activity over a given period, we can identify co-rental patterns that the current analysis does not capture. For example, the temporal data shows Horror spikes in certain months. A basket analysis would reveal whether customers renting Horror are also consistently pulling Thriller, and whether that pairing is stronger in specific months.

Business application

If Horror and Thriller are frequently co-rented during October, the business should stock both heading into that period rather than loading up on Horror alone. The same logic applies to any seasonal pattern. The basket tells you what to stock alongside the headline category, not just which category tops the chart.

Beyond seasonal stocking, co-rental patterns inform in-store layout, promotional bundling, and staff recommendations. A customer asking for a Sports title is more likely to take a second film if the recommendation is drawn from what Sports customers typically rent alongside it.

Why it is relevant

The data to support this analysis already exists in the rental table. Every customer has a rental history that can be grouped by time window and cross-referenced by category. The analysis requires no new data collection, only a different lens on what we already have. Combined with the seasonal trends already identified, it would turn a single-category insight into a stocking and promotion strategy.

6. Recommendations

1. Review pricing on high-volume, low-revenue categories	Action, Family, and Children rank in the top 5 by rental volume but sit in the bottom half by revenue. A modest price adjustment on these categories could close the gap without reducing demand, given their demonstrated popularity.
2. Target retention efforts at high-CLV customers showing churn signals	The 441 customers classified as churned include many who were previously high-frequency renters. Prioritising outreach to customers in the high spend segment who have not rented recently offers the clearest return on retention spend.
3. Redistribute stock from overstocked to understocked titles	Several titles carry 8 copies with fewer than 3 rentals per copy, while high-demand titles have only 2 copies with 5 rentals per copy. Reallocating copies between these titles improves utilisation at no additional cost.
4. Investigate what is driving the store performance gap	Store 2 outperforms Store 1 by 198 rentals but shares a similar category mix. The driver is likely outside the dataset. A review of foot traffic, opening hours, or location factors would clarify whether the gap can be closed or replicated.
5. Commission a market basket analysis before the next stocking cycle	The seasonal data shows which categories peak each month. A basket analysis would reveal what customers rent alongside those categories, enabling smarter stocking and promotional decisions ahead of peak periods.

Technical Appendix

Data sources

The analysis uses the DVD rental PostgreSQL dataset, PostgreSQL relational database modelling a physical video rental store. It contains 15 tables covering films, customers, rentals, payments, inventory, stores, staff, and geographic reference data. The dataset was restored into a Docker-hosted PostgreSQL instance and accessed via SQLAlchemy and pandas.

Tools and libraries

Tool	Purpose
Python 3.13	Primary analysis language
pandas	Data manipulation and aggregation
matplotlib / seaborn	Visualisation
scikit-learn	Churn prediction models
SQLAlchemy / psycopg2	Database connection
Streamlit	Dashboard
Docker / Docker Compose	Infrastructure

Methodology notes

- Customer segmentation uses `pd.qcut` to split customers into three equal-sized buckets by total spend.
- CLV is calculated as average purchase value x purchase frequency x customer lifespan in years.
- Churn thresholds (30 days active, 90 days churned) are set relative to the dataset snapshot date, not the current date.
- Turnover rate is total rentals divided by number of copies per film.
- Payment rate is total payments divided by total rentals per customer.
- Category revenue uses only rentals with a recorded payment. Category volume uses all rentals.

Limitations

- The dataset covers a short observation window with 599 customers and rental activity concentrated between May and August 2005. This limits the depth of any behavioural analysis, makes it difficult to draw concrete business recommendations with confidence, and is the primary reason the churn prediction model cannot generalise beyond the data it was trained on.
- The dataset is a historical snapshot. Churn classifications reflect activity relative to the last recorded date, not today.
- The churn prediction model achieves 1.00 accuracy across all CV folds due to feature-label overlap in a small dataset (599 customers). It is a methodological demonstration, not a deployable model.
- Late fees cannot be isolated from base rental charges in the payment data.

Bias and Ethical Considerations

- The churn model trained on a small, historically skewed dataset may not generalise; the 90-day churn threshold is an assumption that could misclassify customers unfairly
- Using rental behaviour to target customers for retention campaigns raises questions about how that data is used and whether customers are aware their behaviour is being profiled